AN EMPIRICAL STUDY OF MAMBA

A comparison with Transformers under noise and performance analysis across robustness, efficiency, and distribution shifts

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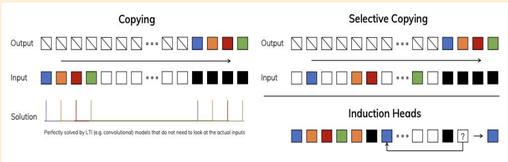
INTRODUCTION



The paper *"Mamba: Linear-Time Sequence Modeling with Selective State Spaces"* introduces a new architecture, Mamba. **Mamba is a proposed sequence model that aims to replace the attention mechanism used in Transformers.** Instead of comparing every token with others, Mamba maintains a running memory that is updated step by step, which allows it to process long sequences with linear time complexity. **A key and interesting innovation of Mamba is its selective mechanism**, where parameters such as Delta and C depend on the input, enabling the model to decide what information to keep, forget, or use for prediction.

CHALLENGES

- One of the important challenges for our project was selecting appropriate **datasets** for evaluation.
- We also faced challenges in the **implementation and training of the simplified Mamba** variants for the ablation study.

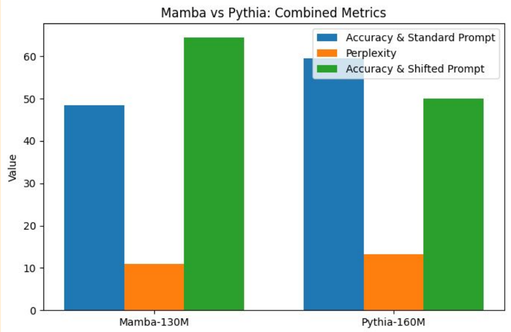
RESEARCH

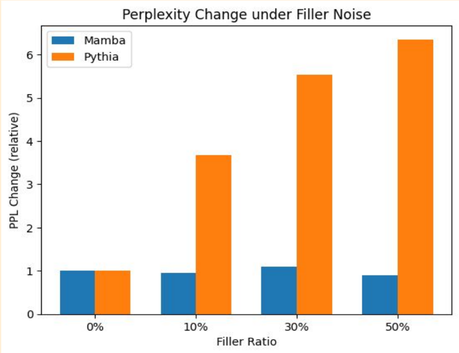
METHODOLOGY

Claim 1: By comparing with baseline, Mamba have a greater advantage in a long sequence (512, 1024, 2048 and 4096)
Claim 2: Mamba-130M achieves higher average accuracy than Pythia-160M in zero-shot evaluations.
Hypothesis 1: Mamba will be more robust than a Transformer when irrelevant filler text is inserted into the input, because Mamba's selective memory should learn to ingore unimportant tokens.
Hypothesis 2: The combination of Selective Delta and Selective C will achieve lower perplexity than using only one of them or removing selectivity.

Dataset:

- PubMedQA (pqa_artificial split): A biomedical question answering dataset sourced from PubMed abstracts.
- WikiText-2 (wikitext-2-raw-v1, test split): A standard English language modeling benchmark derived from Wikipedia articles.





RESULTS

Claim 1: Verified. Mamba increases more slowly compared to the Transformer baseline.
Claim 2: Partially verified. Distribution shift shows that the claim is not suit for some datasets.
Hypothesis 1: Supported. For increasing percentage of filler ratio, there is only small decrease on Mamba's performance even under heavy noise.
Hypothesis 2: Partially supported. The variant that only uses Delta slightly outperforms the full model.

VARIANT	DELTA	C	LOSS	PERPLEXITY
Full_Delta_C	Yes	Yes	7.322	1513
Delta_only	Yes	No	7.311	1496
C_only	No	Yes	7.386	1613
No_Delta_No_C	No	No	7.384	1610

CONCLUSION

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Overall, this project provided us the chance to learn about Mamba and interesting ideas about how **modern sequence models operate beyond the basic Transformer**. Initially, we approached Mamba as simply a more efficient alternative to Transformers, but through experimentation, it became clear that its advantages are not limited to computational efficiency. Also, through the ablation study, we observed that removing these components leads to a noticeable degradation in performance, especially in terms of perplexity.

FUTURE WORK

This project still has some limitations. For our future works and better understanding our hypothesis and Mamba, we would investigate:

- More diverse benchmarks,
- More different datasets,
- Larger-scale training,
- Multiple lengthes sequence for Selective test